TUB-navi (TU Berlin navigation)

A Wiki-style project page - Sustain.ALL Transformation HUB, TU Berlin, Summer Semester 2026, Challenge 05: Sustain.Mobility

TUB-navi (TU Berlin navigation) is a proposed multipurpose navigation application and physical signage system designed to solve campus wayfinding, accessibility and mobility-data communication problems at Technische Universität Berlin. The concept was developed by the Mobility Challenge team of the Sustain.ALL Transformation HUB during the Summer Semester 2026 and prototyped during the Sustain.ALL Future Transformation Hackathon on 1–2 July 2026 at the Einstein Center Digital Future (ECDF), TU Berlin. TUB-navi combines an indoor/outdoor navigation app with upgraded bilingual physical signage to close the data-communication gap between TU Berlin's departments and its more than 43,000 students, staff and visitors.



Figure 1: App logo. Own work.

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Project definition

TUB-navi is the name of the app aiming to ease campus navigation for its users. It was designed by the Sustain.ALL Mobility Challenge team over the course of the SoSe 2026 semester and refined during the two-day Sustain.ALL Digital Transformation Hackathon in July 2026. The underlying idea is based on solutions proposed by the mobility challenge team of the Sustain.ALL HUB at TU Berlin, developed under the guidance of Bengisu Berispek (Project Lead / Lecturer), with the challenge originally scoped together with members from the TU Berlin Mobility Department and the TU Berlin Occupational Safety Department, and led throughout the semester by master's students from TU Berlin's Sustainable Mobility Management programme and HU Berlin's Global Change Geography programme.

Project partners and team

The core semester-long project team consisted of two students who scoped the challenge, conducted the research, and authored the project proposal, working directly with the TU Berlin challenge partners:

Name	Affiliation / Programme	Role
Liska Wohlgemuth	M.Sc. Global Change Geography, HU Berlin	Core team - research, stakeholder mapping, proposal
Santhosh Manjunath	MBA Sustainable Mobility Management, TU Berlin	Core team - research, stakeholder mapping, proposal
Gloria Karcher	TU Berlin Mobility Manager	Challenge partner
Michael Hüllenkrämer	AGUM Coordinator & Occupational Safety, TU Berlin	Challenge partner

At the Future Transformation Hackathon (1–2 July 2026, ECDF), four additional participants joined the team specifically for the hackathon and were responsible for building the first functional app prototype:

Name	Affiliation / Programme	Hackathon contribution
Tammo Snell	B.Sc. Information Systems, TU Berlin	App prototype development
Lennard Sdrojek	M.Sc. Mechanical Engineering, TU Berlin	App prototype development
Peiwen Xie	Student, Lise-Meitner-Schule, Berlin	App prototype development
Simon Mader	Senior Mobility Expert, Deutsche Energie-Agentur GmbH (dena)	Mentorship and mobility-sector expertise

This hackathon team of four - recruited on-site from participants with backgrounds in information systems, mechanical engineering, and secondary-school-level coding interest, together with mentorship from a senior mobility expert at dena - turned the semester-long concept into a working, demonstrable app prototype within the two-day hackathon window.

Project content and goals

Content

TU Berlin's navigation problem rests on five main points: missing data, poor communication, outdated infrastructure, the lack of a single digital tool collecting all relevant information, and the absence of technical maintenance for existing orientation aids.

Goals

The goal of the challenge is to make TU Berlin more accessible, user-friendly and climate-conscious. This is pursued by introducing a centralized navigation system, communicating accessibility information (e.g. wheelchair-approved entrances, elevator status), and supporting a mobility-friendly journey through integration of sustainable transport modes such as public transport connections, EV charging options, and secured bike parking and repair stations. In addition to the digital solution, improvements to physical signage and inter-departmental data sharing are project goals.

The project aligns with four Sustainable Development Goals (SDGs):

SDG (UNSD Indicator)	Target	TUB-navi Contribution
4.a (C040a01)	Disability-, child- and gender-sensitive, inclusive learning environments	Wheelchair routing, real-time elevator alerts, multilingual interface, all-gender toilet signage
9.4 (C090401)	Upgrade infrastructure with clean, resource-efficient technologies	Digital and physical upgrade using existing infrastructure at low cost
11.2 (C110201)	Safe, affordable, accessible, sustainable transport for all	Core mission: campus-wide accessible navigation for 43,000+ users
13.3 (C200306)	Awareness-raising and capacity on climate change	Promotes cycling, walking and e-mobility as default campus transport

Design inspiration for the app's features was drawn from the National University of Singapore (NUS) app and the Technical University of Munich (TUM) campus app. Competitor analysis confirmed that campus navigation apps are a proven, feasible concept; TUB-navi differentiates itself by being TU Berlin-specific, bilingual, accessible, and built around real-time data.



Figure 2: Relevant SDG goals. Source: United Nations Sustainable Development Goals, n.d.

Research and survey insights

A TU Berlin Campus Mobility Survey was conducted online in July 2026, consisting of 36 questions (open answer, multiple choice, single choice) and shared via university networks, social media, e-mail and word of mouth. 20 of 21 started response sheets were completed.

Key findings:

- More than 75% of respondents currently rely on Google Maps for on-campus navigation, despite it lacking indoor data for TU Berlin.
- Existing physical signage frequently fails first-time visitors and is reported as outdated and German-only.
- 95% of respondents expressed interest in a dedicated, TU Berlin-specific navigation app; zero respondents gave a flat “no” to using a free TU Berlin navigation app.

From these findings, five intervention points were defined: QR codes on existing signage linking to the digital app; bilingual signage at key decision points (entrances, gates, complex-building entrances, main intersections); improved data sharing between TU Berlin departments; a maintenance and user-reporting feedback loop; and integration of the navigation aids into student orientation and onboarding.

Solution concept

TUB-navi is a dual-layer navigation ecosystem consisting of a digital and a physical layer that reinforce one another: physical signs make the app discoverable, while the app supplies real-time data that static signage alone can never provide.

Digital layer - TUB-navi app (free, bilingual German/English)

- Indoor maps and building-entrance locators - filling the gap Google Maps does not cover
- Real-time elevator, door-opener and accessibility status for mobility-impaired users
- Live parking and EV / e-bike charging availability
- Wheelchair-accessible and cycling-friendly route planning
- QR-code scanning from physical signs for instant live-map activation
- BVG / public-transport integration for multi-modal, sustainable commuting
- One-tap user reporting of broken barriers, incorrect signage, or accessibility failures
- Emergency indoor-routing mode for emergency services

Physical layer - upgraded signage

- Bilingual (German + English) directional signage at 5-10 highest-traffic decision points
- QR codes embedded into existing signs, upgrading current infrastructure at near-zero cost
- Accessibility-focused tactile markers and wheelchair route indicators at key entrances

The system is designed to track, over time: infrastructure (roads, pedestrian crossings, emergency muster points, emergency-vehicle access, public-transport connection schedules); buildings and rooms (cafeteria, cafés, pack stations, elevators, automatic doors, indoor mapping); entrances including wheelchair accessibility; bathrooms including showers and gender-free options; drinking wells; bike and car parking including weather-protected and secured options; lockers; charging points for e-bikes, EVs and phones; and bike-repair stations.

Business Model Canvas

A Business Model Canvas (Strategyzer format) was worked through by hand during the Research & Pilots phase (May-June 2026) to test the viability of TUB-navi / TUni(B) across key partners, activities, resources, value propositions, customer relationships, channels, customer segments, cost structure and revenue streams - captured on-site by Liska Wohlgemuth.

Block	Notes
Key Partners	TU Berlin (different departments) · Berliner Senat / Bezirksamt (cooperation on mobility) · Technical partners - external operations partners
Key Activities	App data collection, storage, indication of signs & maps (physically) · Research & analysis · Design & development · Monitoring & maintenance
Key Resources	Mobile development · survey data · tips from students · Maps & data for TU management, logistics, urban development
Value Propositions	Easy access · User-friendly navigation (personalized way of mobility) · From TU, for TU & guests
Customer Relationships	App identity, rating for improvement · Openness for input on sudden obstructions/damage, real-time functions at all times · Live data shared & verified by TU logistics when damage occurs
Channels	TU IS / social media / on physical maps (QR code) · As part of welcome info sent by mail
Customer Segments	Students & TU employees · Visitors & guests · Delivery drivers · Emergency responders
Cost Structure	Build-in / development costs · Land/space · Sign board
Revenue Streams	Minimal share from semester fees (not extra cost for students) · Revenue from delivery services / bike-rental credit within campus · Optional collaboration with delivery services

Figure: Business Model Canvas worked through by the team during the Research & Pilots phase, May–June 2026. Photo: L. Wohlgemuth.

Hackathon and prototype development

The Sustain.ALL Future Transformation Hackathon took place on 1-2 July 2026 at the Einstein Center Digital Future (ECDF), TU Berlin, bringing together interdisciplinary teams from across the Sustain.ALL HUB to co-create prototypes and strategies for their respective sustainability challenges, supported by workshops and mentorship, ahead of a final pitch to a jury and audience on 15 July 2026.

For Challenge 05 (Sustain.Mobility), four additional participants - Tammo Snell (B.Sc. Information Systems, TU Berlin), Lennard Sdrojek (M.Sc. Mechanical Engineering, TU Berlin), Peiwen Xie (student, Lise-Meitner-Schule, Berlin), and mentor Simon Mader (Senior Mobility Expert, dena) - joined the core Liska/Santhosh team specifically for the two hackathon days. Working from the semester's research, survey insights, and solution concept, this hackathon sub-team built the first functional TUB-navi app prototype using publicly accessible data from TU Berlin, OpenStreetMap, and the open-source TUM Campus App GitHub repository as a technical reference and data foundation.

The prototype demonstrated the core navigation concept validated by the survey - TU Berlin-specific indoor and outdoor routing - and served as the centrepiece of the team's pitch at the Future Transformation Showcase on 15 July 2026, where all Sustain.ALL HUB challenge teams presented their results to a jury and audience.



Figure 3: Screenshot of the app prototype interface. Created during the Hackathon, July 2026.

Application scope / implementation

The current implementation status of TUB-navi is a research-validated concept plus an early hackathon prototype: indoor/outdoor navigation logic and a basic map interface were demonstrated using open TU Berlin and OpenStreetMap data, referencing the architecture of the open-source TUM Campus App. Full implementation would require: securing a student or external developer team; sourcing live data feeds for parking, elevators and EV/bike charging from TU Berlin's facilities and IT departments; a GDPR compliance review for any location or usage data collected; and coordination with TU Berlin's Campus Administration and IT Department as data and infrastructure gatekeepers. The application is intended to be shareable across universities, with a design that could combine campus navigation with related sustainability features such as building-level energy information or a mensa food-tracking system to support healthier, more environmentally conscious habits.

Funding

Cost estimates

Cost category	SoSe 2026 (pilots)	WiSe 2026 (development)
Survey tools	€0 (Google Forms)	-
Pilot materials (kiosk, printing, rewards)	€200–500	-
Bilingual sign prototypes (5–10 locations)	€300–800	-
App development - student developer team	-	€5,000–15,000
App development - external vendor	-	€30,000–50,000
GDPR compliance review	-	€1,000–3,000
Contingency (15%)	~€75–200	~€5,000–8,000

Sources of finance

Potential funding pathways identified for TUB-navi include the BMBF MobilitätsWerkStadt 2025 programme (following the precedent set by the related MobCC project), TU Berlin's institutional sustainability funds, partnership grants from the Bezirksamt Charlottenburg-Wilmersdorf, resources from the Neue Mobilität Berlin (NMB) consortium, and Berlin Senate Mobility Act implementation grants. No funding has been secured to date; this remains an open item for the project's next phase.

Results and further development

The app concept is based on existing university campus apps, TU Berlin's current situation and needs as defined by the challenge partners, and the positive results of the Campus Mobility Survey conducted in July 2026 (75%+ current reliance on Google Maps; 95% interest in a dedicated TU Berlin app; 0% flat refusals). The Hackathon (1–2 July 2026) turned this validated concept into a working prototype, presented at the Future Transformation Showcase on 15 July 2026.

TUB-navi connects policy, campus administration, IT and mobility teams with end users to improve navigation through one integrated app and physical signage system. A continuous feedback loop is intended to ensure that user needs and campus updates keep improving the service, ultimately supporting more than 35,000 students, staff and visitors in navigating TU Berlin easily, safely and sustainably.

Challenges and lessons learned

Over the course of the semester and the hackathon, the team encountered and worked through the following challenges:

- Data collection - key mobility, accessibility and infrastructure data is siloed across TU Berlin departments and was not readily accessible.
- Stakeholder mapping - identifying and reaching the right contacts across a large, decentralized institution (TU Mobility, IT, Campus Administration, Safety) required significant coordination effort.
- User survey - recruiting sufficient respondents within a short survey window was harder than anticipated (20 of 21 started responses completed).
- Time constraints - the two-day hackathon format required rapid prioritization to move from concept to demonstrable prototype.
- Changes in planning - the project scope and prototype plan were adjusted several times as new stakeholder input and hackathon team capabilities became clear.
- App prototype development - building a working navigation prototype in two days required relying on open data sources (OpenStreetMap, TU Berlin public data, TUM Campus App code) rather than TU Berlin's own unreleased internal data.
- Communication and cultural differences - coordinating a team spanning two universities, multiple degree programmes, a secondary-school participant, and an industry mentor required adapting communication styles and expectations.

Next steps

- Design a refined prototype in Figma, incorporating insights from the competitor analysis and user requirements gathered so far.
- Develop the prototype into a functional mobile application with a dedicated developer team.
- Launch a pilot version for TU Berlin students and staff.
- Collect user feedback via usability testing, surveys and usage analytics.
- Iterate and improve based on pilot feedback.

Three supporting initiatives are proposed to further promote adoption:

Initiative	Purpose / mechanism	Expected outcome
Bike-to-Campus Challenge	4-week gamified challenge (leaderboards, rewards) at semester start	Data on cycling habits and app feature requests
Pop-Up Mobility Kiosk	Temporary kiosk with bike repair, QR codes to the app, physical signage samples	Insights on navigation pain points and feature priorities
In-app trip logging	Participants log trips for points and real-time tracking (e.g. parking availability)	Ongoing engagement and usage data

References

Karcher, G., & Center for Research on Environmental Decisions in Berlin (CREDiBl). (2026). TU Berlin Campus Mobility Survey (Q1 2026) [Unpublished raw data]. Technical University of Berlin.

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National University of Singapore. (n.d.). NUS Campus Map. https://www.nus.edu.sg/campusmap/ (last accessed 07 Aug 2026).

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Weblinks

Google survey link, Mobility challenge (2026)

Sustain.ALL Transformation HUB

TU Berlin — campustransformation symposium and publication

TUM Campus App (GitHub)

NUS Campus Map

MobCC project, TU Berlin

Neue Mobilität Berlin (NMB)

Note: The TU Berlin Campus Mobility Survey (Q1 2026) referenced above is treated as confidential and should not be published or shared with third parties.